26.02.03

MC : $\langle S, P \rangle$: $\Pr[S_{t+1} = s' \mid S_t = s]$

$$P_{ss'} = \Pr[S_{t+1} = s' \mid S_t = s, S_{t-1}, S_{t-2} \dots S_1]$$

MRP $\langle S, P, R, \gamma \rangle$ R_s ; $G_t = R_t + \gamma R_{t+1} + \gamma^2 R_{t+2} + \dots$

$$v(s) = \mathbb{E}[G_t \mid S_t = s]$$

$$v(s) = R_s + \gamma \cdot P_{s,s'} v(s')$$

MDP $\langle S, P, R, \gamma, A \rangle$ $\pi(a|s) \rightsquigarrow A_t \mid S_t = s$

$$v_\pi(s) = E_\pi[G_t \mid S_t = s], \quad q_\pi(s,a) = E_\pi[G_t \mid S_t = s, A_t = a]$$

$$\Rightarrow v_\pi(s) = \sum_a \pi(a|s) q_\pi(s,a), \quad q_\pi(s,a) = R_s^a + \gamma \cdot \sum_{s'} P_{ss'}^a v_\pi(s')$$

$$v_*(s) = \max_\pi v_\pi(s) \qquad v_*(s) = \max_a q_*(s,a)$$

$$q_*(s,a) = \max_\pi q_*(s,a) \qquad q_*(s,a) = R_s^a + \gamma \cdot \sum_{s'} P_{ss'}^a v_*(s')$$

MC - Evaluation

1 2 3 ...... T

Episode 1 : $S_1 A_1 R_1 \cdot S_2 A_2 R_2$ ...... $S_T A_T R_T$ π

Episode 2 : $S_1 A_1 R_1$ ...... $S_T A_T R_T$ π

$V_\pi(s)$ ① First - Visit

(i) Episode -1 $S_5 = s$ $G_5 = R_5 + \gamma \cdot R_6 + \dots$

$\gamma^{T-5} \cdot R_T$

(ii) For all episodes compute G

(iii) Average of all collected G's

② Every - Visit :

Episode -1 $S_5 = s$, $S_{17} = s$

Collect both G_5 & G_{17}